

M5.1 Unit description

Applications of vectors in mechanics; variable mass; moments of inertia of a rigid body; rotation of a rigid body about a fixed smooth axis.

M5.2 Assessment information

Prerequisites

A knowledge of the specifications for M1, M2, M3 and M4 and their prerequisites and associated formulae, together with a knowledge of scalar and vector products, and of differential equations as specified in FP2, is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

Work done by a constant force = $\mathbf{F} \cdot \mathbf{d}$

Angular momentum = $I\dot{\theta}$

Rotational kinetic energy = $\frac{1}{2}I\dot{\theta}^2$

$L = I\ddot{\theta}$

$$\int_{t_1}^{t_2} L \, dt = I\omega_2 - I\omega_1$$

For constant angular acceleration:

$$\omega_1 = \omega_0 + \alpha t$$

$$\omega_1^2 = \omega_0^2 + 2\alpha\theta$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = \frac{(\omega_0 + \omega_1)}{2} t$$

1 Applications of vectors in mechanics

What students need to learn:

Solution of simple vector differential equations.

Vector differential equations such as

$$\frac{d\mathbf{v}}{dt} = k\mathbf{v},$$

$$\frac{d^2\mathbf{r}}{dt^2} + 2k\frac{d\mathbf{r}}{dt} + (k^2 + n^2)\mathbf{r} = \mathbf{f}(t)$$

where k and n are constants,

$$\frac{d\mathbf{r}}{dt} + 4\mathbf{r} = \mathbf{ie}'.$$

Use of an integrating factor may be required.

Work done by a constant force using a scalar product.

Moment of a force using a vector product.

The moment of a force $\mathbf{F}$ about O is defined as $\mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is the position vector of the point of application of $\mathbf{F}$.

The analysis of simple systems of forces in three dimensions acting on a rigid body.

The reduction of a system of forces acting on a body to a single force, single couple or a couple and a force acting through a stated point.

2 Variable mass

What students need to learn:

Motion of a particle with varying mass.

Students may be required to derive an equation of motion from first principles by considering the change in momentum over a small time interval.

3 Moments of inertia of a rigid body

What students need to learn:

Moments of inertia and radius of gyration of standard and composite bodies.

Use of integration including the proof of the standard results given in the formulae booklet will be required.

The parallel and perpendicular axes theorems.

4 Rotation of a rigid body about a fixed smooth axis

What students need to learn:

Motion of a rigid body about a fixed smooth horizontal or vertical axis.

Use of conservation of energy will be required. Calculation of the force on the axis will be required.

Angular momentum.

Kinetic energy.

Conservation of angular momentum. The effect of an impulse on a rigid body which is free to rotate about a fixed axis.

Simple pendulum and compound pendulum.